

FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-113 Final Production Gate

Protected Course 3 graduation assessment: run, diagnose, recover, change, revalidate and defend

Status	RC1 - FINAL-LAYER QA PASS CANDIDATE
Course boundary	C3-104 = C3-09 phase Gate only; C3-113 = separate course-wide graduation Gate
Release boundary	Course 3 is NOT FINAL until Mentor/PWA + whole-course release regression are complete.

Permanent engineering behavior controls the course. Vendor UI/version notes are refreshable references, and external links are support-only; the course artifacts remain sufficient without them.

1. Graduation boundary

C3-104 proves C3-09 phase mastery. C3-113 is a separate whole-course Gate. Passing C3-104, completing all lessons, or earning an optional certification does not substitute for C3-113.

2. Preconditions

- All ten phase Gates are PASS under their competency rules.
- RetailPulse satisfies C3-110 architecture acceptance criteria and C3-111 source contracts.
- README, ADR, runbook, environment/version record and evidence index are reproducible.
- Live-runtime evidence exists for every platform/runtime capability the learner claims to have executed.
- C3-112 remains sealed from the learner and a fresh protected variant is available.
- During the active Gate: no AI solving/debugging/architecture-answer generation, no protected review, and no tutorial-solution search.

3. Five required events

A - Clean run	Run declared inputs to trusted serving output and prove baseline reconciliation.
B - Unseen incident	Receive one fresh protected defect/failure and diagnose it from quality/operational evidence.
C - Recovery	Contain, repair and replay/backfill only the necessary boundary without duplicate/corrupt business effects.
D - Controlled change	Implement one fresh schema/business/config/access/SLO change with versioned contract/test updates.
E - Revalidation + defense	Rerun tests/reconciliation and independently defend architecture, security, limitations, cost/performance and 10x scaling.

4. Strict PASS rule

PASS requires ALL of the following: every one of the eight competencies is marked PASS; all five required events are complete; every required evidence category is present and reproducible; the protected variant is fresh; architecture defense is independent; and there is zero unresolved critical failure. There is no overall percentage, average, points total, or compensating strength in another competency.

5. Eight competencies - each stands alone

Correctness & requirements	Meets declared behavior and change request without silently changing the contract.
Data integrity & validation	Independent key/count/value controls prove trusted state before and after change/recovery.
Reliability & recovery	Retry/replay/backfill is bounded, idempotent and does not duplicate side effects.
Testing & change safety	Critical tests fail when they should; controlled change cannot bypass required validation.
Observability & diagnosis	Signals identify the failing boundary and support an evidence-based diagnosis.
Security & governance	Secret/privacy/access boundaries are correctly implemented and evidenced.
Maintainability & reproducibility	Reviewer can recreate the required run from saved materials and documented versions.
Design reasoning & ownership	Learner independently explains trade-offs, limitations, failure modes and scale changes.

6. Required evidence packet

- Clean run ID + input hashes/versions + successful reconciliation.
- Incident signal + diagnosis notes + containment boundary.
- Recovery/replay/backfill evidence + reconciled trusted result.
- Versioned controlled-change request + implementation + tests + contract/runbook update.
- Post-change tests/reconciliation and unchanged critical invariants where applicable.
- Reproduction instructions + environment/runtime version evidence.
- Independent architecture-defense notes recorded by reviewer.

7. Critical automatic failures

- Publish or defend unreconciled data corruption as trusted.
- Expose a secret or sensitive field in repository/log/public evidence.
- Fabricate tests, logs, runtime evidence or reviewer evidence.
- Unsafe recovery creates duplicate/corrupt business state.
- Depend on undocumented hidden/manual state that prevents reproduction.
- Use AI/protected-answer/tutorial-solution assistance during the active Gate.

8. Retest routing

A failed final attempt is retained. Targeted repair may address the weak competency, but graduation still requires a fresh full final attempt using a different protected variant/change request. A successful earlier phase Gate never substitutes for this retest.